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TASK	WINDOWS FORENSICS 1 REPORT

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WINDOWS FORENSICS 1 REPORT

Introduction

In this module, I explored the fundamentals of Windows digital forensics, focusing on identifying and analyzing key system artifacts. I learned how to use forensic tools to investigate user activity, system changes, and potential security incidents.

1.0 Introduction to Windows Forensics

I learned that computer forensics involves collecting and analyzing digital evidence to understand user activity on a system. I understood that Windows systems store forensic artifacts, which are traces of user actions, in various locations like the registry and user directories. I also realized that these artifacts are created to improve user experience but can be used by investigators to reconstruct events.

Question: What is the most used Desktop Operating System right now?
Answer: **Microsoft Windows**

2.0 Windows Registry and Forensics

I learned that the Windows Registry is a structured database that stores configuration data about the system, users, and applications. I understood that it is organized into keys and values, with five main root keys that help categorize different types of information. I also learned that the registry is a valuable source of forensic artifacts used to trace user activity on a system.

Question: What is the short form for HKEY_LOCAL_MACHINE?
Answer: **HKLM**

3.0 Accessing registry hives offline

I learned how to access Windows registry hives from an offline system by locating their files on disk. I understood the importance of key hives like SAM, SYSTEM, and NTUSER.DAT, as well as specialized ones like Amcache for tracking executed programs. I also learned that transaction logs and backups can provide additional forensic evidence, including recent or previously deleted data.

Question1: What is the path for the five main registry hives, DEFAULT, SAM, SECURITY, SOFTWARE, and SYSTEM?
Answer: **C:\Windows\System32\Config**

Question2: What is the path for the AmCache hive?
Answer: **C:\Windows\AppCompat\Programs\Amcache.hve**

4.0 Data acquisition

I learned that data acquisition is the process of creating a copy of system data to ensure forensic analysis does not alter the original evidence. I understood that registry hives are protected files and require specialized tools like KAPE, Autopsy, or FTK Imager to extract them. I also learned that different tools provide various methods for acquiring registry data from live systems or disk images.

5.0 Exploring Windows Registry

I learned that exported registry hives cannot be viewed using the default Windows Registry Editor and require specialized forensic tools. I understood how tools like Registry Explorer, Registry Viewer, and RegRipper are used to analyze registry data, each with different capabilities. I also learned the importance of incorporating transaction logs to ensure more accurate forensic analysis.

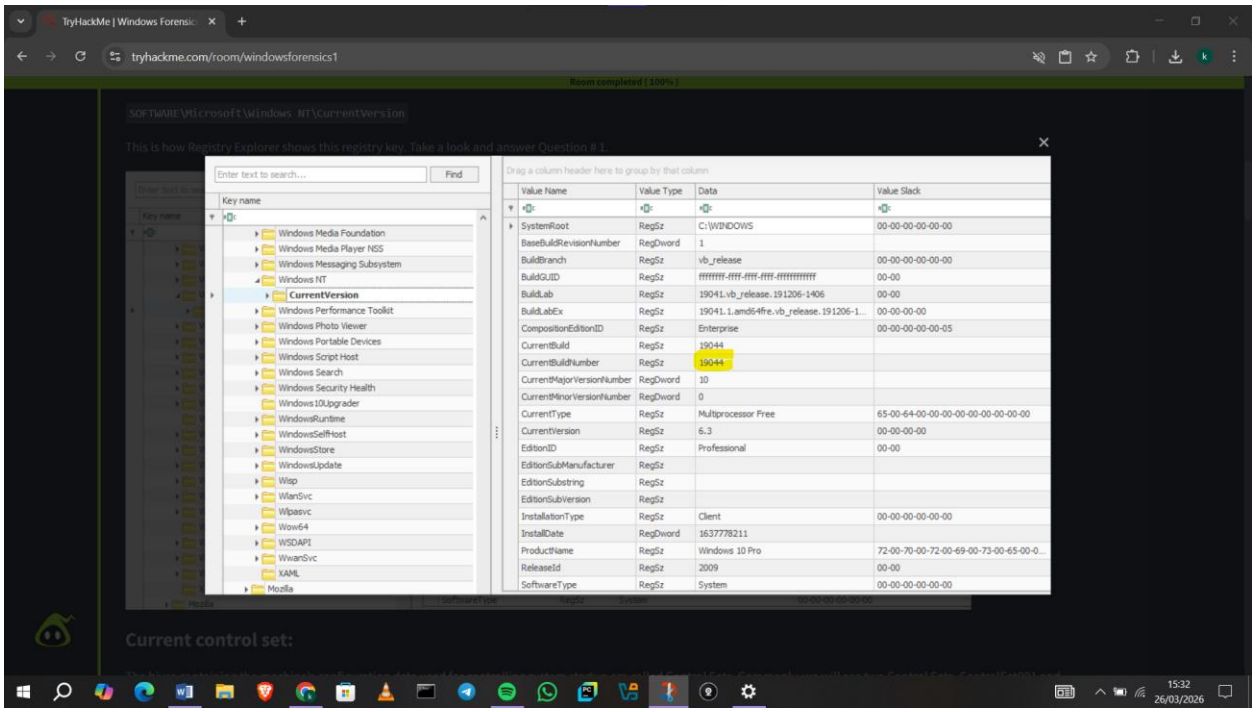
6.0 System Information and System Accounts

I learned how to locate important system and user information within the Windows Registry for forensic analysis. I understood how to identify details such as the OS version, active control set, computer name, time zone, network activity, and autostart programs. I also learned how the SAM hive provides detailed information about user accounts and login activity on the system.

Question: What is the Current Build Number of the machine whose data is being investigated?

Answer: **19044**

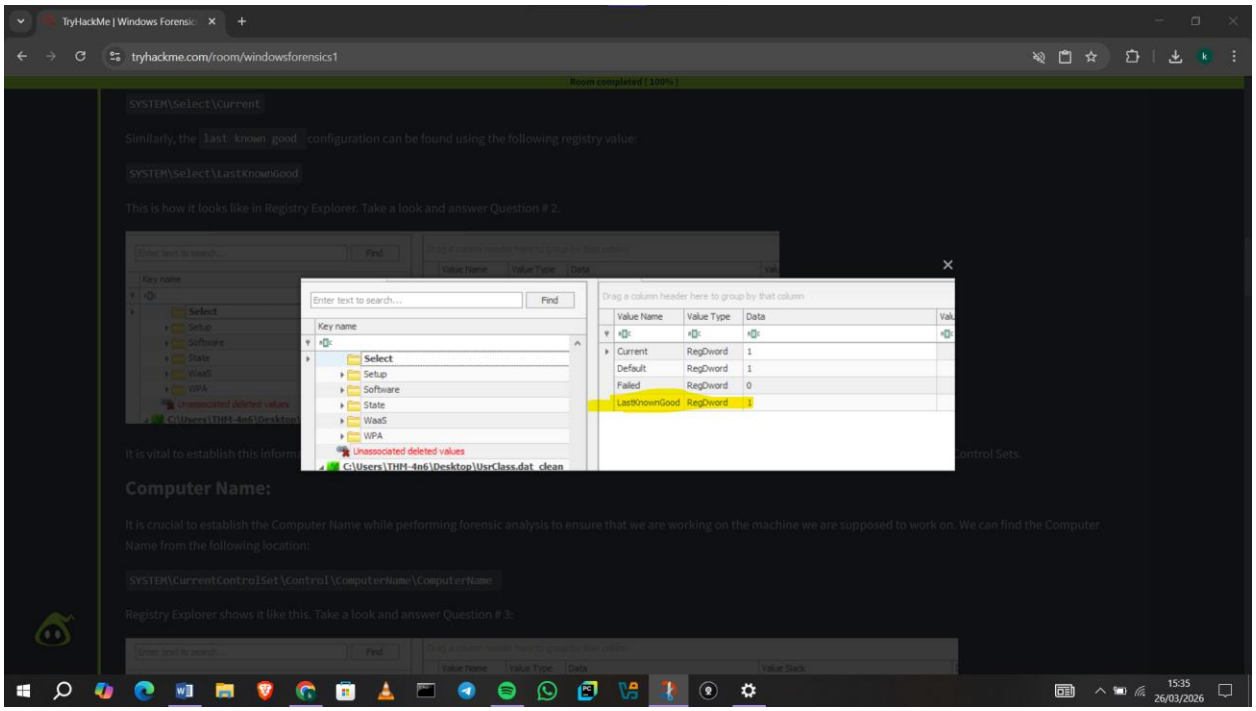
(SOFTWARE\Microsoft\Windows NT\CurrentVersion)



Question: Which ControlSet contains the last known good configuration?

Answer: **1**

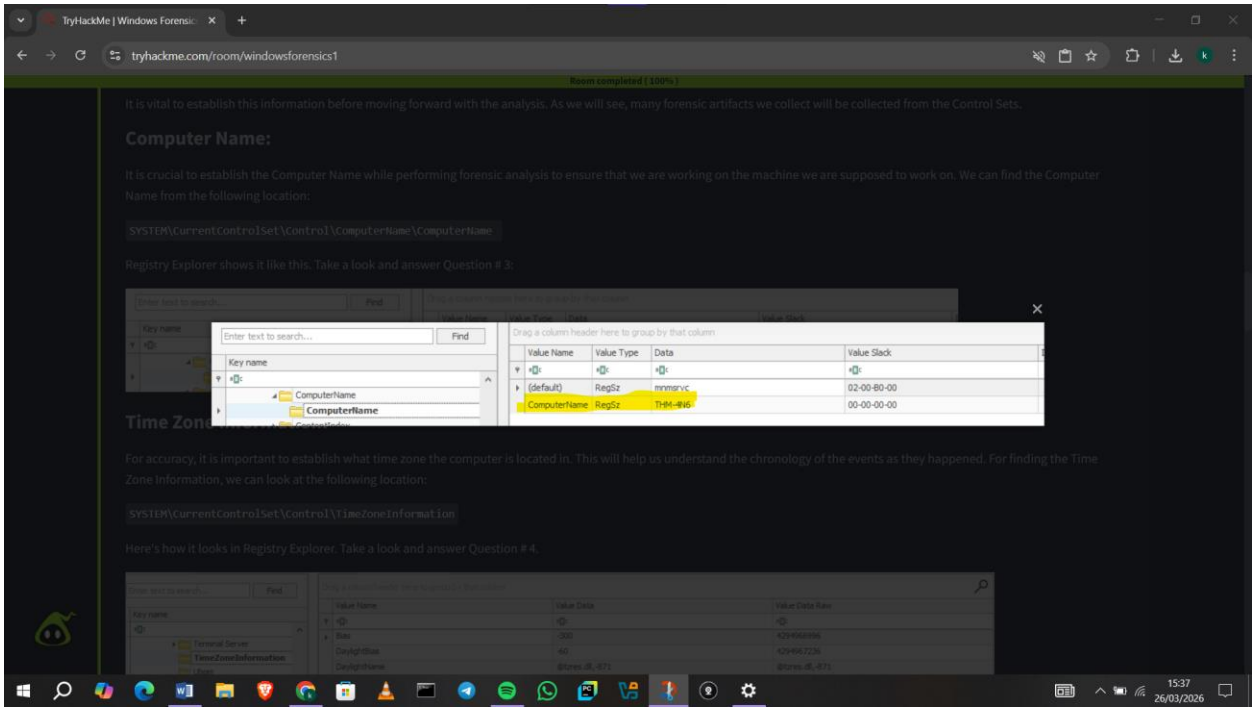
(SYSTEM\Select\LastKnownGood)



Question: What is the Computer Name of the computer?

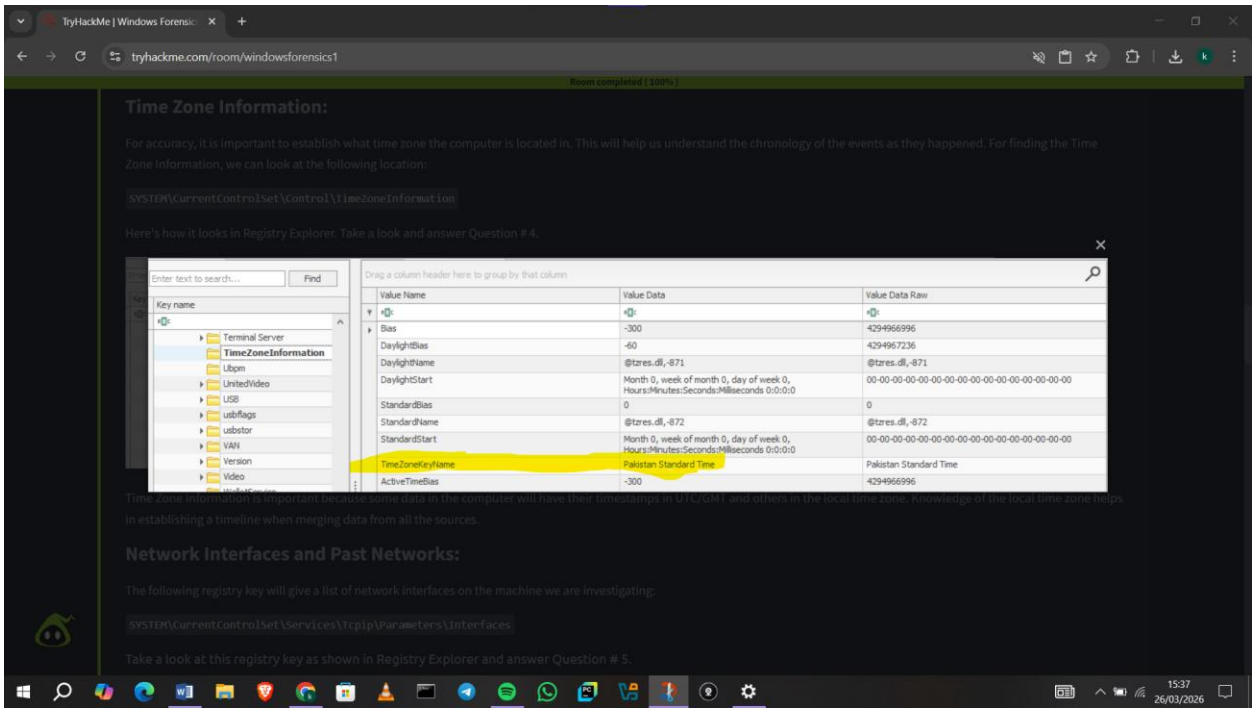
Answer: **THM-4n6**

(SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName)

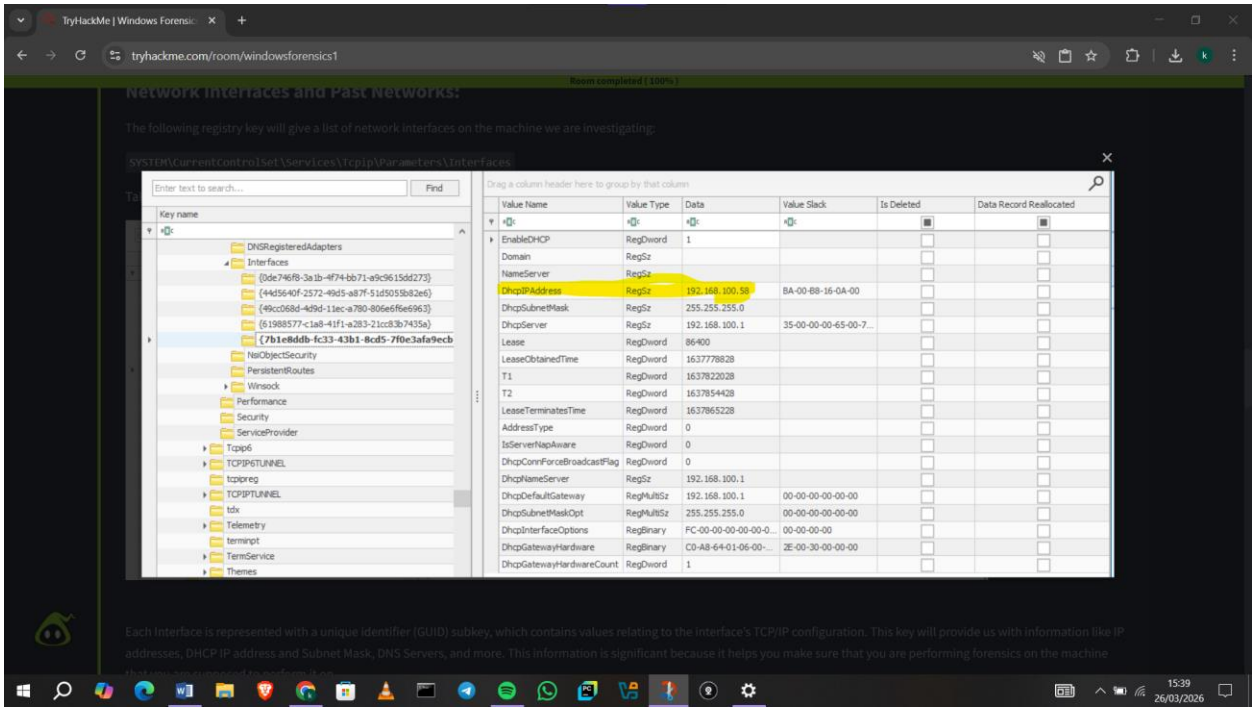


Question: What is the value of the TimeZoneKeyName?

Answer: **Pakistan Standard Time**
(SYSTEM\CurrentControlSet\Control\TimeZoneInformation)



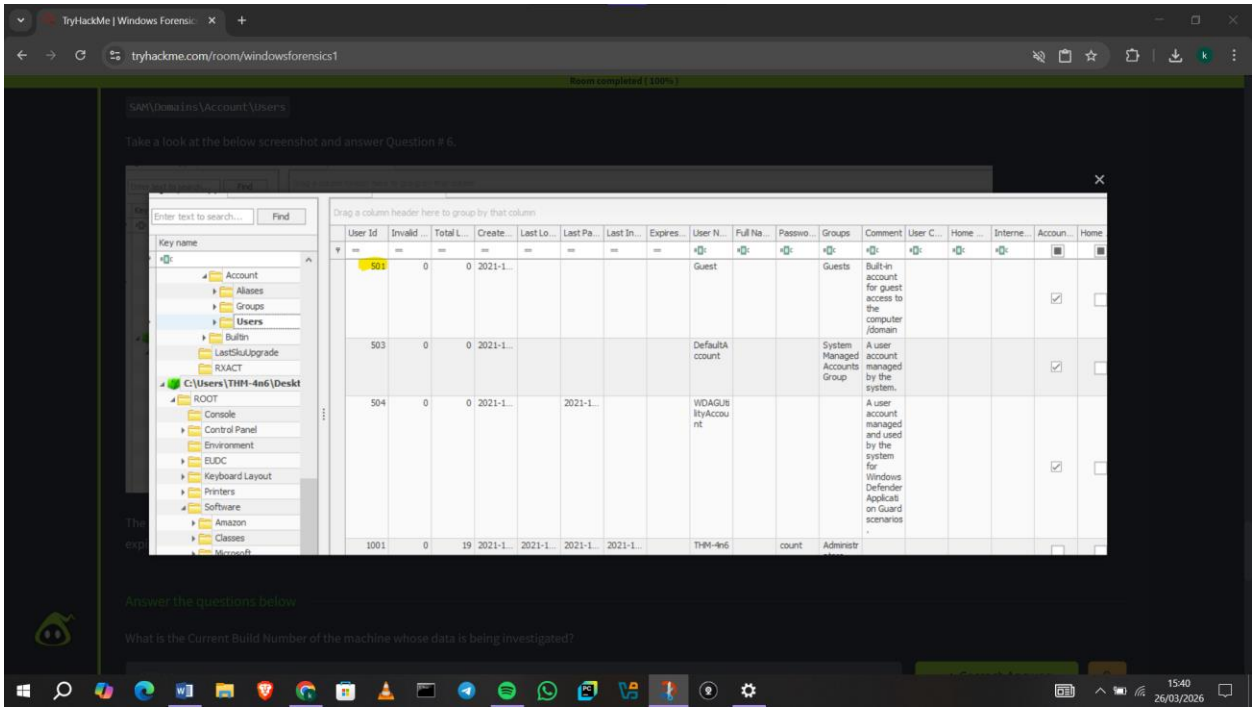
Question: What is the DHCP IP address?
Answer: **192.168.100.58**
(SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces)



Question: What is the RID of the Guest User account?

Answer: **501**

(SAM\Domains\Account\Users)



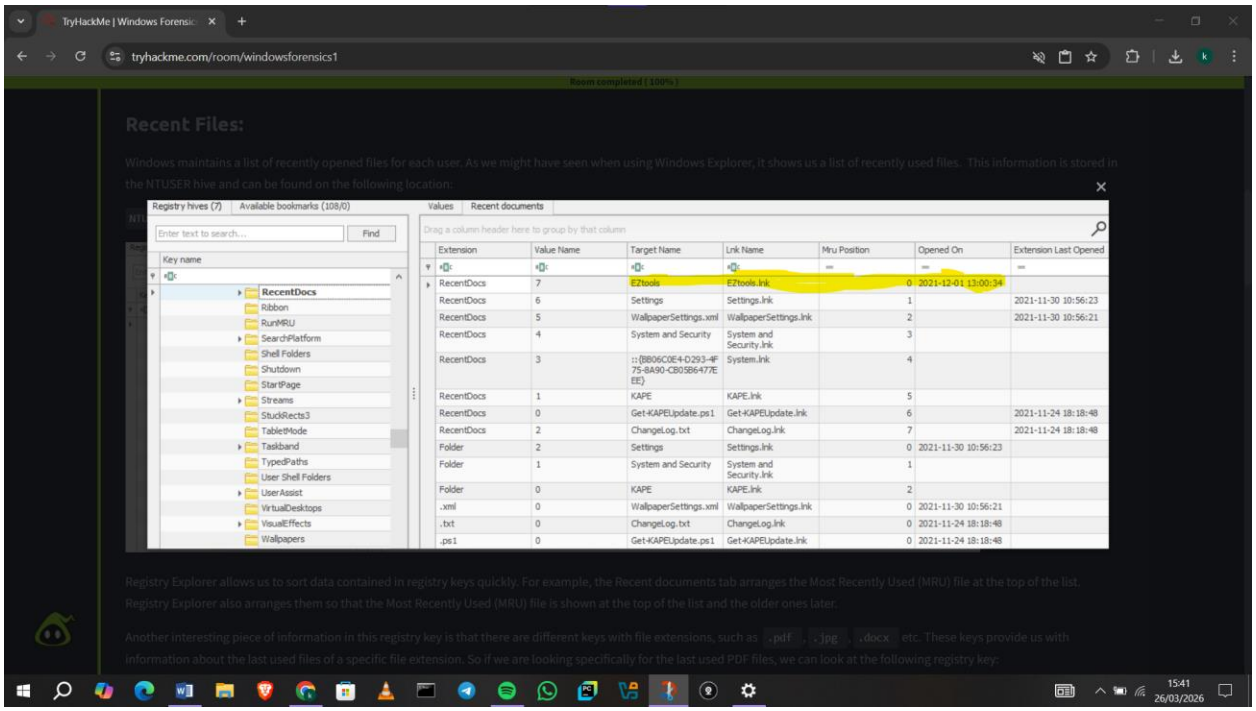
7.0 Usage or knowledge of files/folders

I learned how to identify a user's interaction with files using various registry artifacts such as RecentDocs, Office recent files, and ShellBags. These artifacts can reveal recently opened files, accessed folders, and even deleted file paths. I also learned that additional registry keys can show file access through open/save dialogs and user search or navigation activity.

Question: When was EZtools opened?

Answer: **2021-12-01 13:00:34**

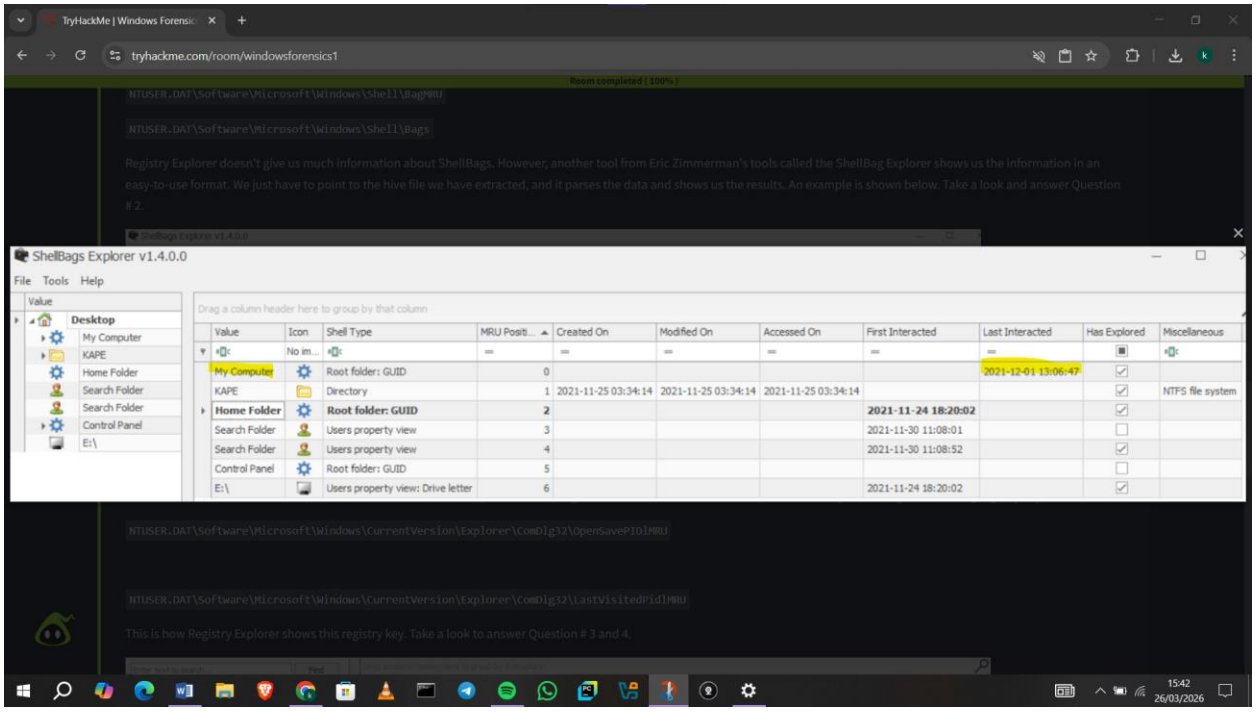
(NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs)



Question: At what time was My Computer last interacted with?

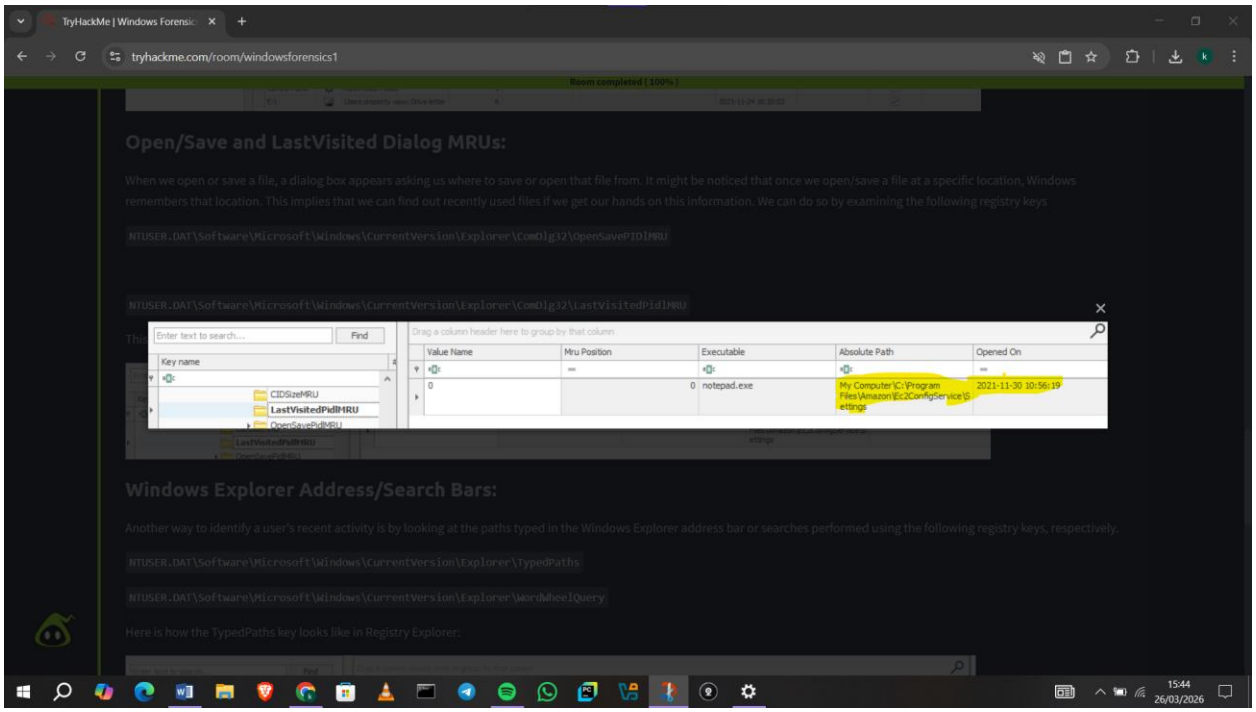
Answer: **2021-12-01 13:06:47**

(USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\Bags)



Question: What is the Absolute Path of the file opened using notepad.exe?

Answer: **C:\Program Files\Amazon\Ec2ConfigService\Settings (NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidlMRU)**

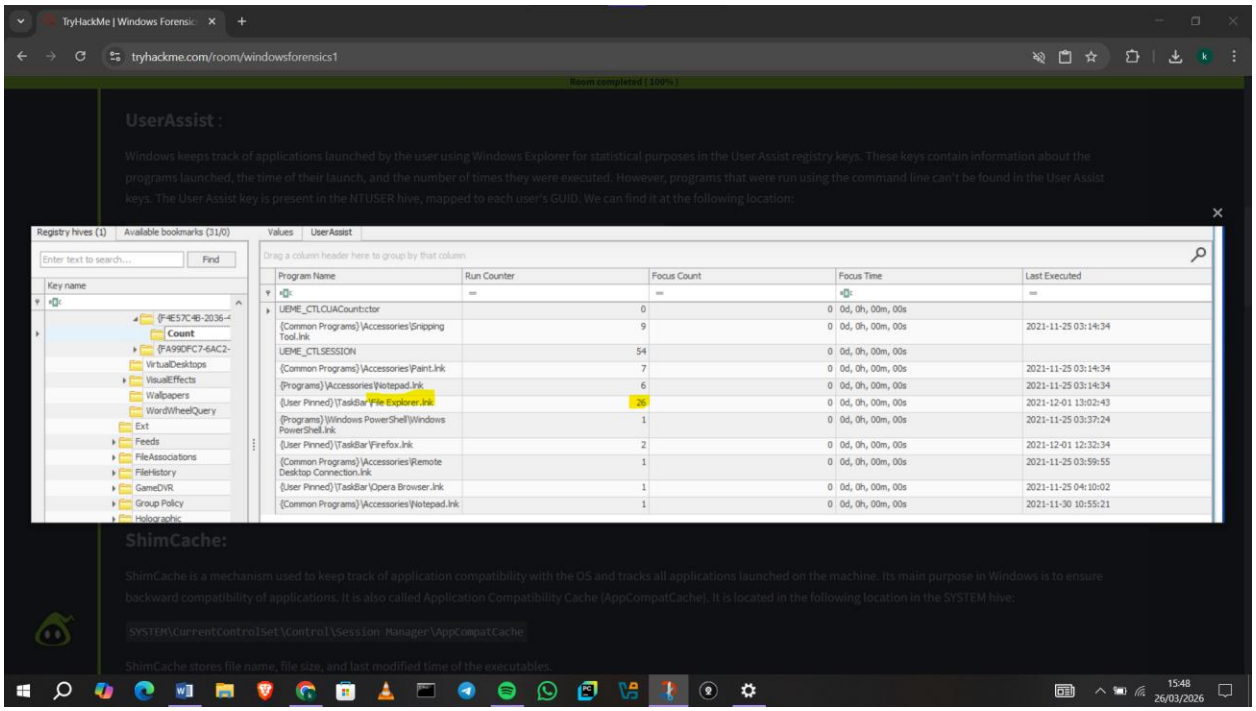


Question: When was this file opened?
Answer: **2021-11-30 10:56:19**

8.0 Evidence of Execution

I learned how to identify evidence of program execution using registry artifacts such as UserAssist, ShimCache, AmCache, and BAM/DAM. They provide different levels of detail, including execution times, file paths, and frequency of use. I also learned that combining multiple sources helps build a more accurate picture of application activity on a system.

Question: How many times was the File Explorer launched?
Answer: **26 (NTUSER.DAT\Software\Microsoft\Windows\Currentversion\Explorer\UserAssist\{GUID}\Count)**



Question: What is another name for ShimCache?
Answer: **AppCompatCache**

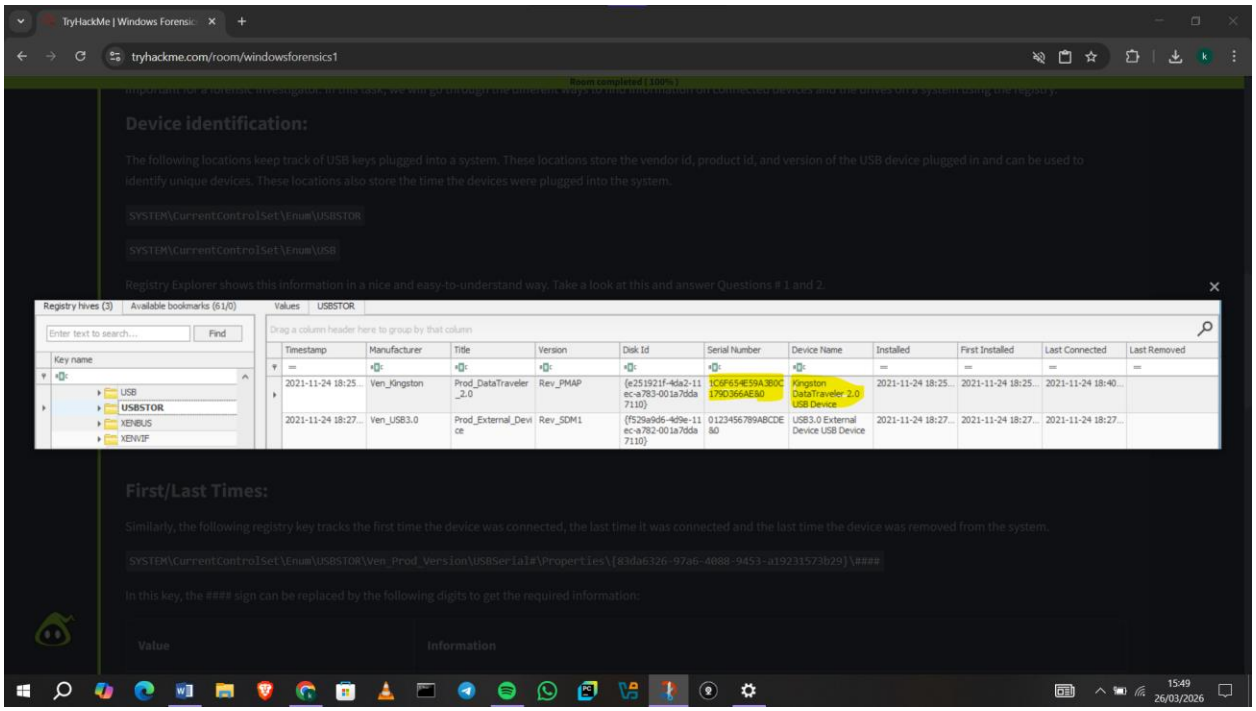
Question: Which of the artifacts also saves SHA1 hashes of the executed programs?
Answer: **AmCache**

Question: Which of the artifacts saves the full path of the executed programs?
Answer: **BAM/DAM**

9.0 External Devices/USB device forensics

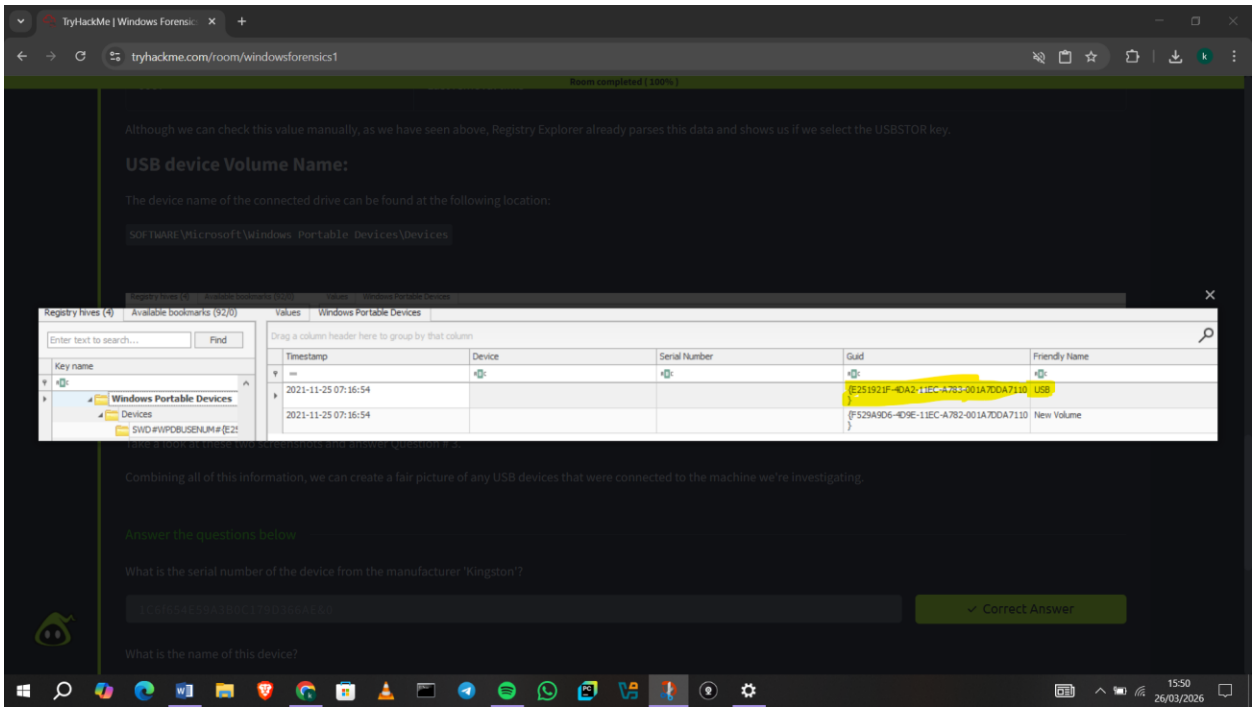
I learned how to identify and analyze USB and removable devices connected to a Windows system using registry artifacts. I understood how to determine device details such as vendor information, connection times, and removal times. I also learned how to correlate device identifiers with volume names to build a complete picture of external device usage.

Question: What is the serial number of the device from the manufacturer 'Kingston'?
Answer: **1C6f654E59A3B0C179D366AE&0**
(SYSTEM\CurrentControlSet\Enum\USBSTOR)



Question: What is the name of this device?
Answer: **Kingston Data Traveler 2.0 USB Device**

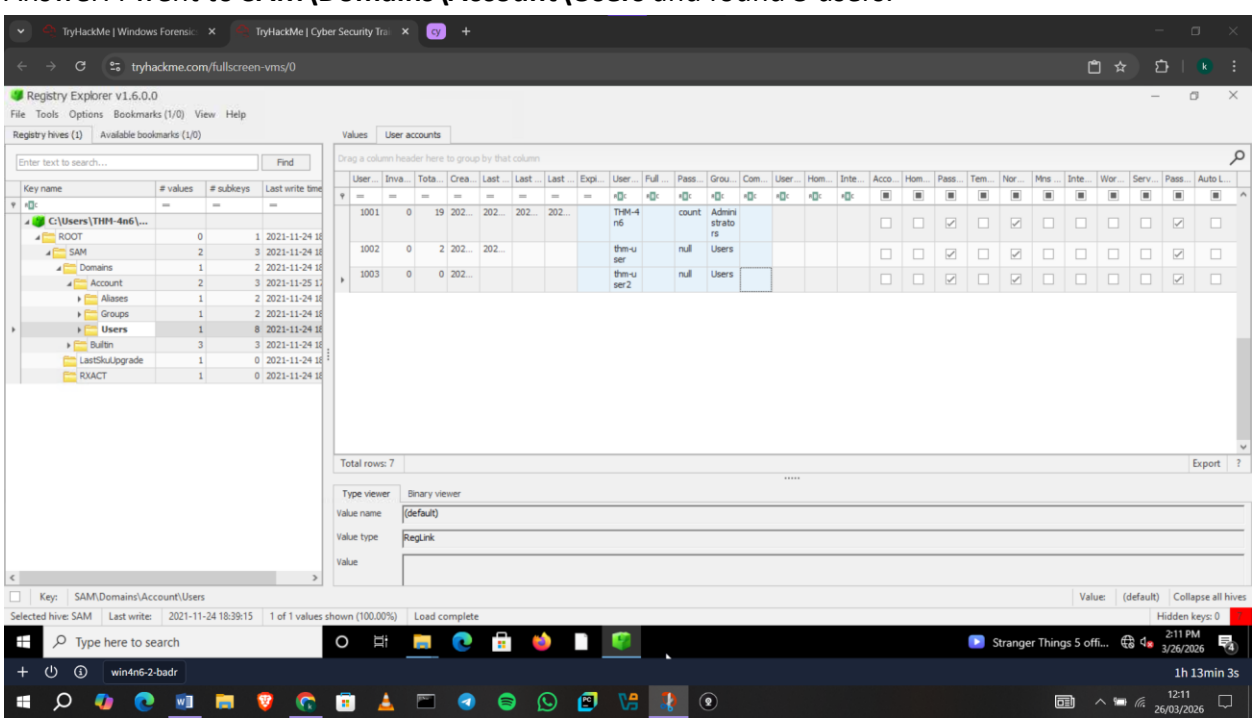
Question: What is the friendly name of the device from the manufacturer 'Kingston'?
Answer: **USB**
(SOFTWARE\Microsoft\Windows Portable Devices\Devices)



10.0 Hands-on Challenge

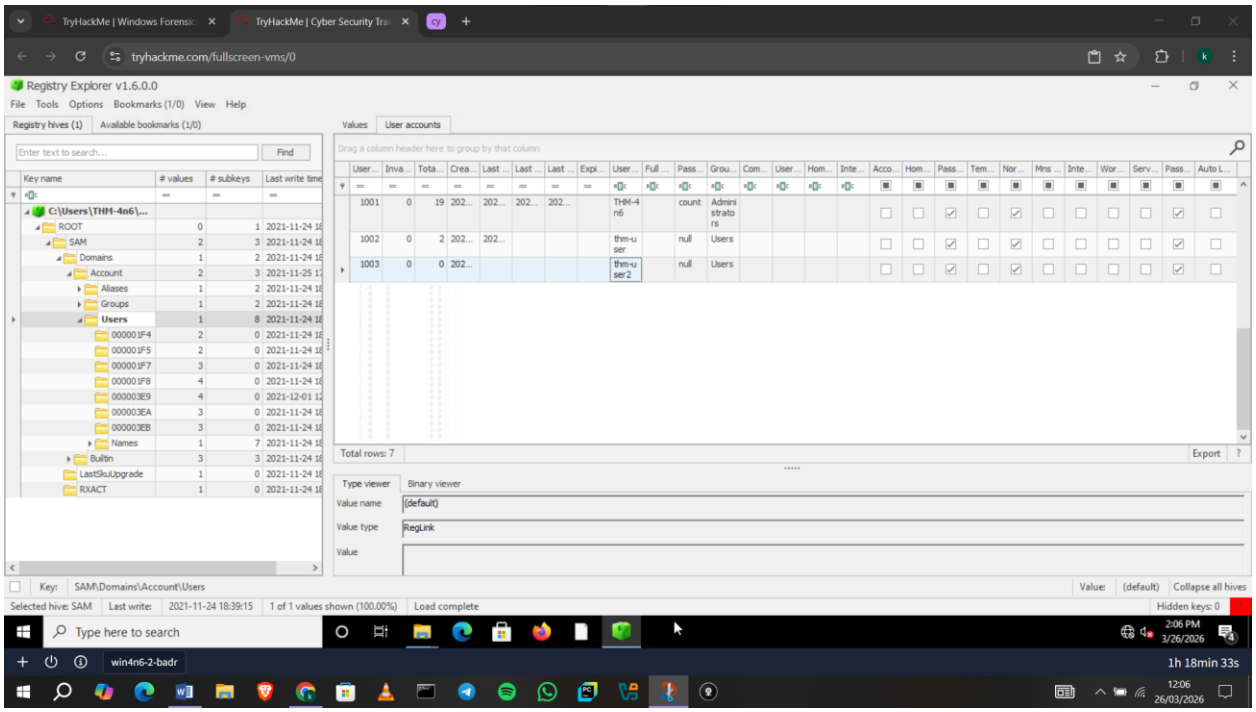
Question: How many user created accounts are present on the system?

Answer: I went to **SAM\Domains\Account\Users** and found **3** users.



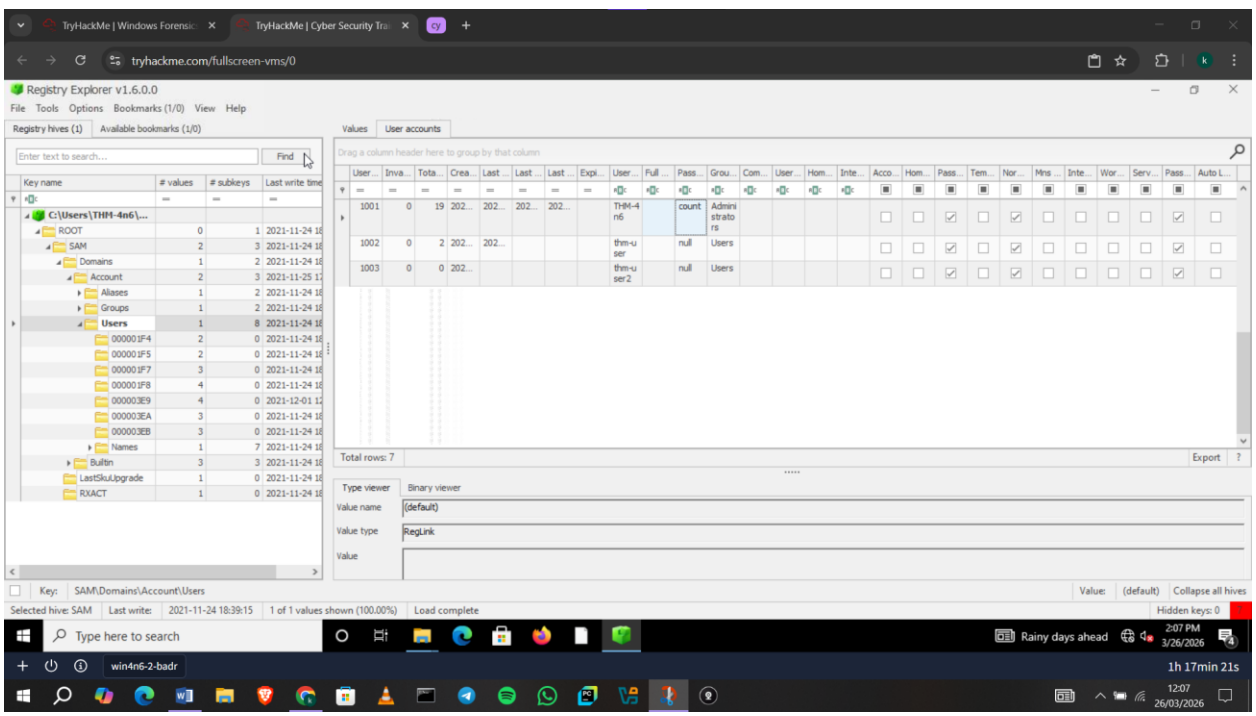
Question: What is the username of the account that has never been logged in?

Answer: I found **thm-user2** with 0 login count at **SAM\Domains\Account\Users.**



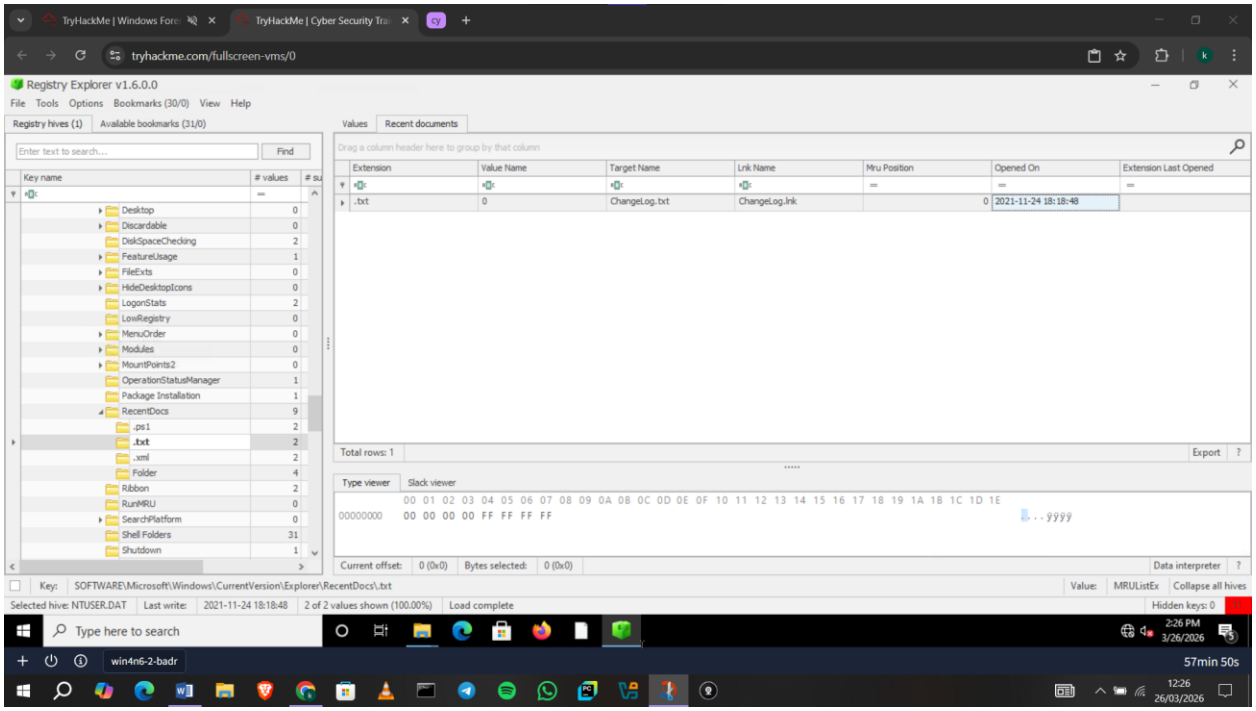
Question: What's the password hint for the user THM-4n6?

Answer: **count**



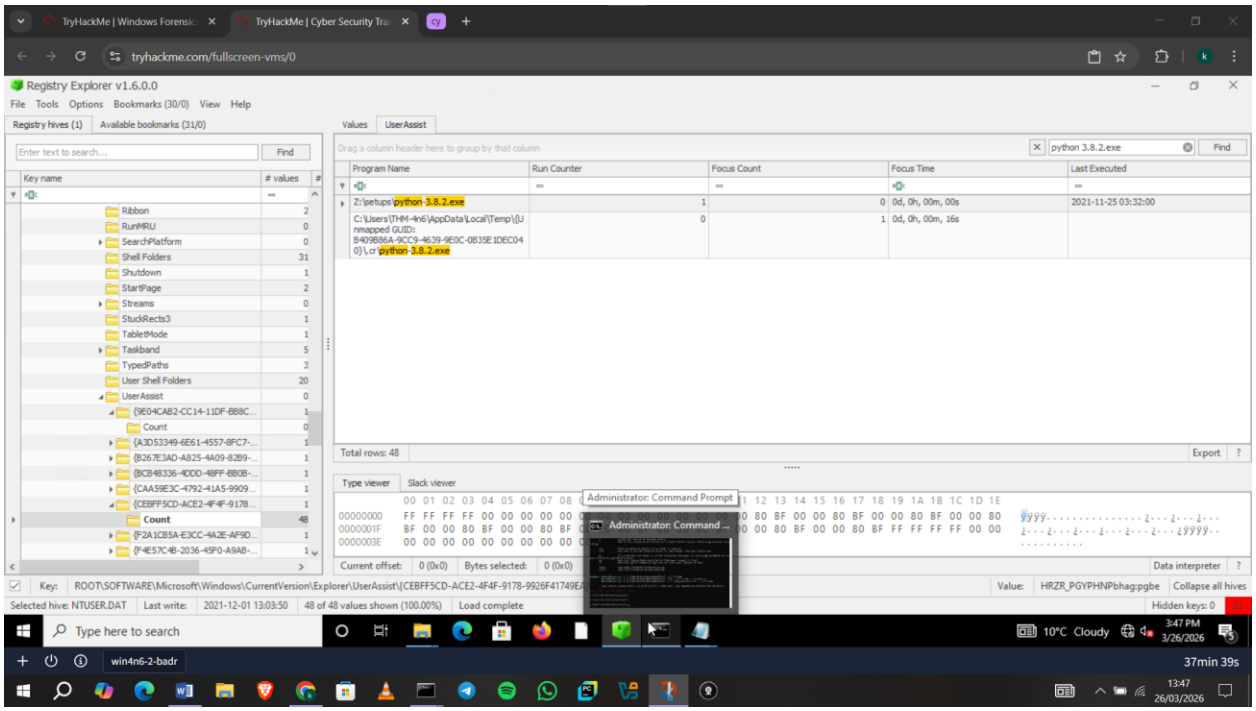
Question: When was the file 'Changelog.txt' accessed?

Answer: I found the file at **NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs** and found the last date it was opened as **2021-11-24 18:18:48**.



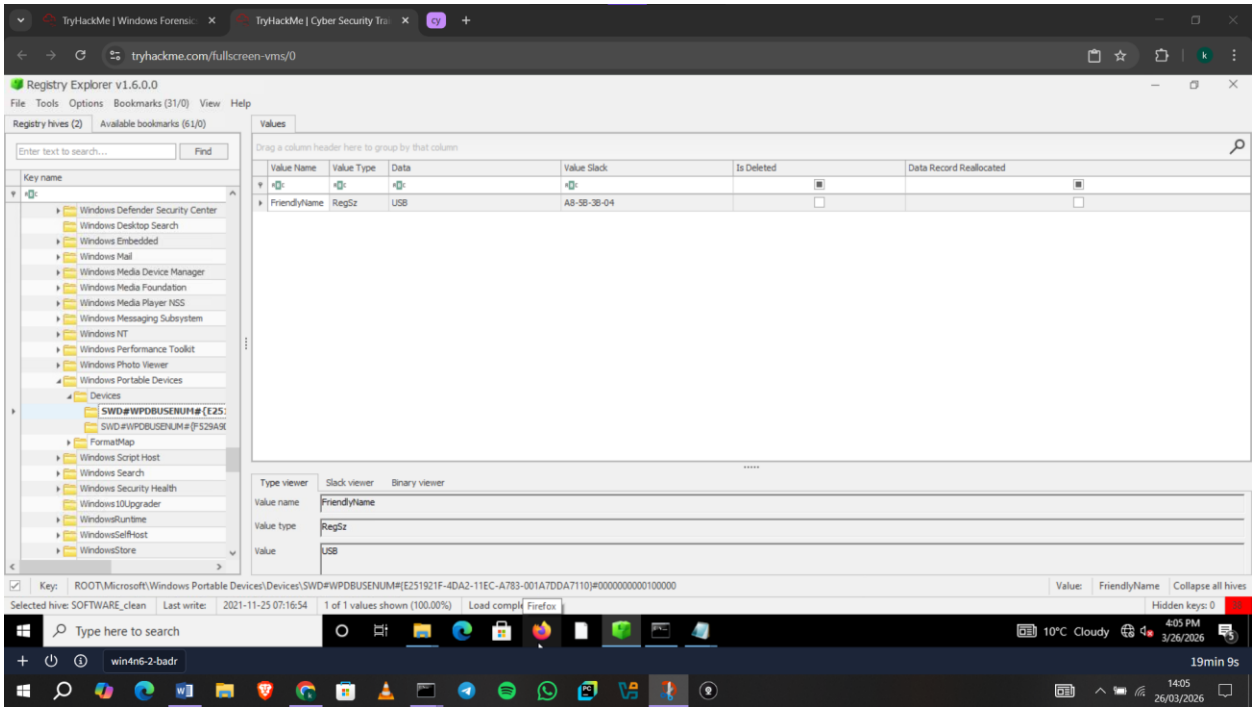
Question: What is the complete path from where the python 3.8.2 installer was run?

Answer: I went to **NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist** and found the path: **Z:\setups\python-3.8.2.exe**.

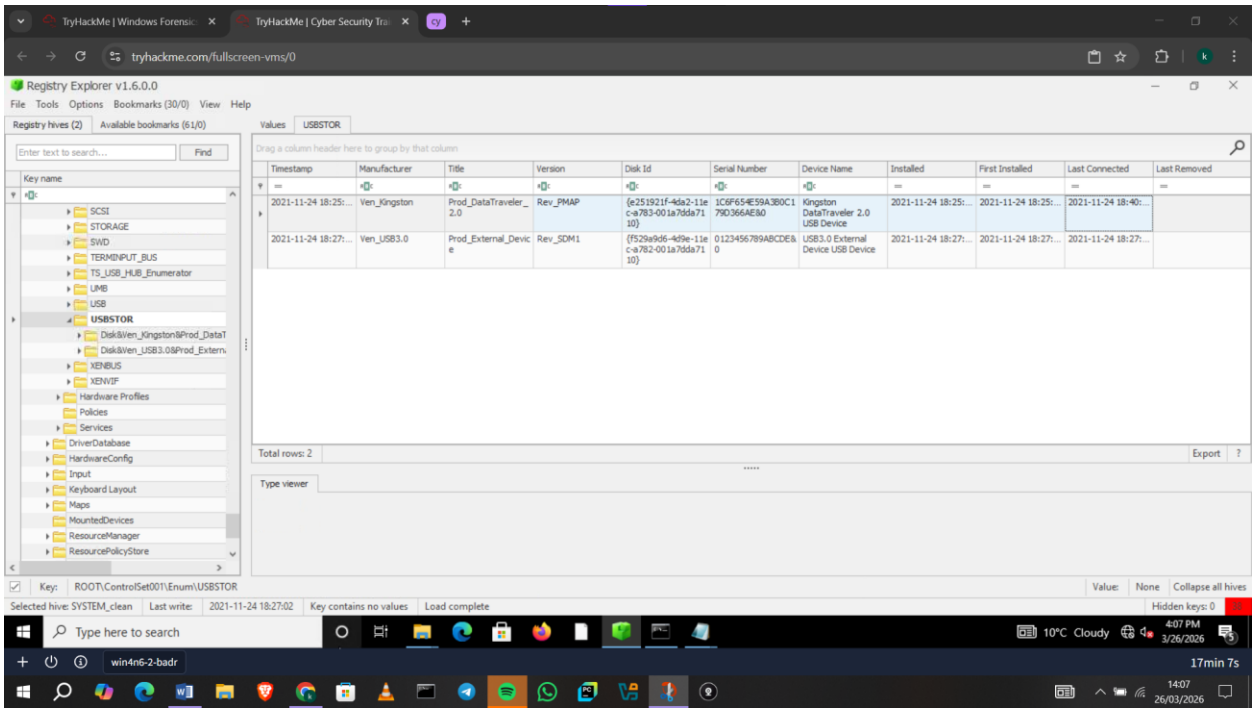


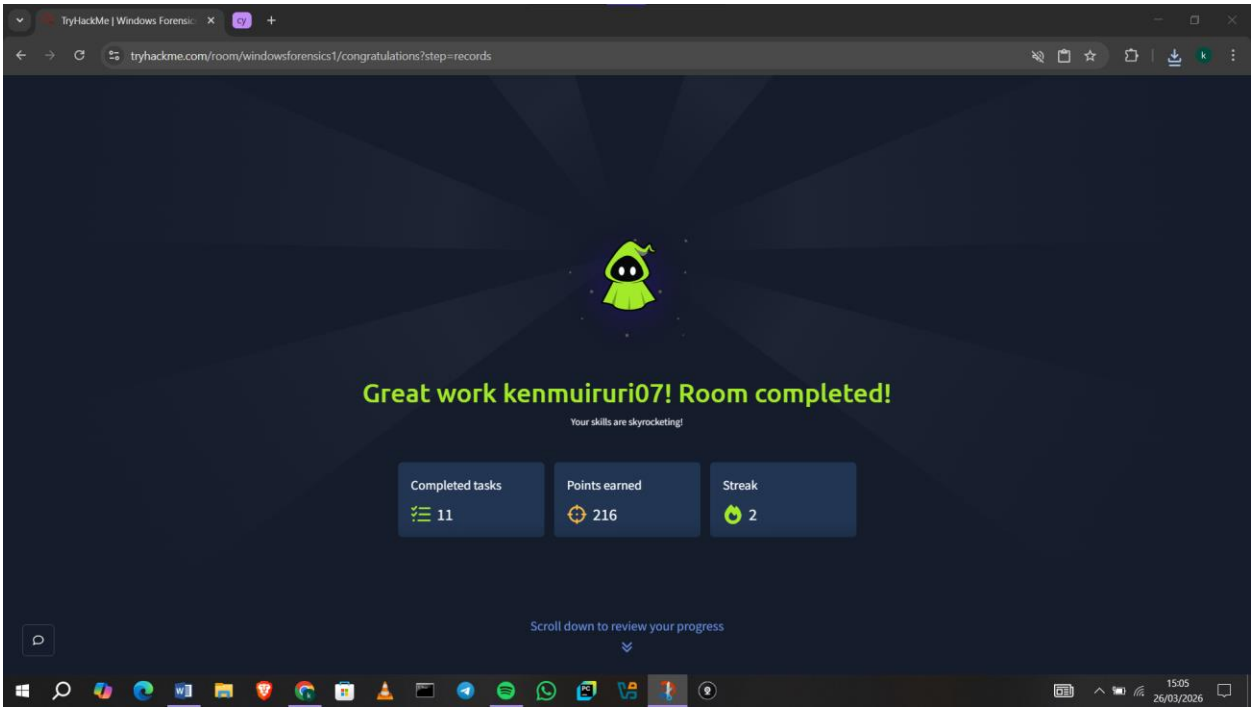
Question: When was the USB device with the friendly name 'USB' last connected?

Answer: I went to **SOFTWARE\Microsoft\Windows Portable Devices\Devices** and noted the **Disk ID** for USB.



I then went to **SYSTEM\CurrentControlSet\Enum\USBSTOR** and found the Disk ID for USB and the last time it was connected was **2021-11-24 18:40:06**.





Conclusion

Overall, this module strengthened my understanding of how digital evidence is stored and analyzed within Windows systems. I am now more confident in using forensic techniques and tools to reconstruct events and support investigations.